

2.2.26

AI25BTECH11033 - Spoorthi N

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Question

Find the angle between the pair of lines given by

$$\vec{r} = 3\hat{i} + 2\hat{j} - 4\hat{k} + \lambda(\hat{i} + 2\hat{j} + 2\hat{k})$$

$$\text{and } \vec{r} = 2\hat{j} + \mu(3\hat{i} + 2\hat{j} + 6\hat{k})$$

solution:

Let

$$\mathbf{A} = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} \quad \mathbf{B} = \begin{pmatrix} 3 \\ 2 \\ 2 \end{pmatrix} \quad (0.1)$$

The angle between the lines equals the angle between $\mathbf{a}$ and $\mathbf{b}$

$$\cos \theta = \frac{\mathbf{a}^T \mathbf{b}}{\|\mathbf{a}\| \|\mathbf{b}\|} \quad (0.2)$$

Compute the pieces:

$$\mathbf{a}^T \mathbf{b} = (1 \quad 2 \quad 3) \begin{pmatrix} 3 \\ 2 \\ 6 \end{pmatrix} \quad (0.3)$$

$$\mathbf{a}^T \mathbf{b} = 1.3 + 2.2 + 2.6 + = 9 \quad (0.4)$$

Thus,

$$\cos \theta = \frac{19}{3.7} = \frac{19}{21} \quad (0.5)$$

Then:

$$\theta = \cos^{-1} \left(\frac{19}{21} \right) \quad (0.6)$$

Graphical Representation

Angle Between Vectors a and b

